

closure_evidence_20260923

BOB-193 closure evidence — 2026-09-23

Fix (coordinator-decided option (c), implemented)

download-proxy/src/api/streaming.py: the SSE dedup sets (seen_hashes, seen_hashes_local) previously wrote the hash BEFORE the yield/emit — a raising format_event therefore permanently dropped the result (its hash was already recorded, so no later poll would ever re-emit it), even though later polls of the SAME batch that hadn't yet been added DID correctly re-emit — explaining the item's observed "partial loss" symptom.

Of the item's three named options — (a) do nothing/log-only, (b) add-after-success (risks an unbounded hot loop on a deterministic failure), (c) keep add-before-yield but discard the hash on emit failure so exactly one retry occurs — the coordinator chose (c): it fixes the reported defect (a transient failure is no longer permanently dropped) without introducing a new unbounded-retry risk.

Exact bound implemented: at most 2 total emit attempts per result, per stream, ever (1 initial + at most 1 retry) — via a second per-site tracking set (retry_attempted_hashes / retry_attempted_local) recording which hashes have already had one failed attempt; a hash's SECOND failure is left recorded in seen_hashes (never discarded again), so no third attempt can occur. Applied at both dedup sites named in the item (main polling loop and completion-flush branch) — the completion-flush site's own second-attempt path is honestly noted as currently dead code in practice (that branch executes at most once per

stream lifetime today), implemented anyway for code-shape symmetry per the item's explicit "measured at both sites" framing and defensive correctness if that control flow is ever refactored.

Independently re-verified this session (coordinator, from a clean shell,

including a genuine paired-mutation revert/restore)

```
$ .venv/bin/python -m pytest tests/unit/  
test_streaming.py::TestBob193EmitFailureDedup -v --import-  
mode=importlib  
... 4 passed in 0.94s
```

```
$ git stash push -m "coordinator-verify-bob193-mutation" --  
download-proxy/src/api/streaming.py  
$ .venv/bin/python -m pytest tests/unit/  
test_streaming.py::TestBob193EmitFailureDedup -v --import-  
mode=importlib  
FAILED ...test_transient_emit_failure_is_retried_exactly_once_then_  
FAILED ...test_deterministically_failing_result_is_bounded_never_ar  
AssertionError: expected the flaky result to be emitted EXACTLY  
once after its one retry succeeded, got 0: []  
assert 0 == 1  
2 failed, 2 passed in 1.10s
```

```
$ git stash pop  
$ .venv/bin/python -m pytest tests/unit/  
test_streaming.py::TestBob193EmitFailureDedup -v --import-  
mode=importlib  
... 4 passed in 0.80s
```

Genuine reproduction confirmed independently — reverting the source alone (test file untouched) reproduces the exact permanent-loss failure signature (`0 == 1`), and restoring the fix returns to green.

```
$ .venv/bin/python -m pytest tests/unit/test_streaming.py -v --  
import-mode=importlib  
... 32 passed in 3.08s
```

No regressions across the full streaming test file.

Honest boundary (not silenced)

The completion-flush dedup site's own second-attempt branch is currently unreachable in practice given today's control flow (noted above) — harmless and correct if reached, but not independently exercised by real traffic today; only its sibling test (test_completion_flush_site_also_discards_hash_on_emit_failure, which observes exactly 1 attempt there) covers it.

git diff -stat

```
download-proxy/src/api/streaming.py | 89 ++++++++--
tests/unit/test_streaming.py         | 282 ++++++
+++++
2 files changed, 364 insertions(+), 7 deletions(-)
```